

Rushi Ashok Savani

AI/ML Engineer | Generative AI | LLM Systems | Agentic AI | MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying Generative AI, Agentic AI, NLP, and production ML solutions across healthcare, enterprise, and retail environments. Expertise in LLMs, RAG, text-to-SQL, LangGraph, PyTorch, AWS SageMaker, MLflow, Docker, Kubernetes, and LLMOps, with a strong focus on scalable and reliable AI systems. Delivered measurable business impact including 98% answer faithfulness, 95% faster data lookups, 24% fewer AI production issues, and ~\$95K in annual savings.

TECHNICAL SKILLS

Languages & Development: Python, SQL, TypeScript, Node.js, Bash, Git, REST APIs, FastAPI, Postman, Jupyter

Machine Learning: Scikit-learn, XGBoost, Gradient Boosting, SVM, Decision Trees, HDBSCAN, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Classification, Regression, A/B Testing, Hypothesis Testing

Deep Learning & NLP: PyTorch, TensorFlow, Hugging Face Transformers, BERT, RoBERTa, Sentence-BERT, Embeddings, Fine-Tuning, Sentiment Analysis

Generative AI & LLM: LLMs, RAG, Agentic AI, Text-to-SQL, Model Context Protocol (MCP), LangChain, LangGraph, LlamaIndex, Google ADK, LiteLLM, Prompt Engineering, Guardrails, RAGAS, LangSmith, LLMops

MLOps & Cloud: AWS, Microsoft Azure, SageMaker, EKS, OpenSearch, DynamoDB, Redshift, CloudWatch, MLflow, Docker, Kubernetes, GitHub Actions, CI/CD, Redis, OpenTelemetry, Model Monitoring

Data & BI: Apache Spark, Pandas, NumPy, SQL Server, Snowflake, pgvector, Qdrant, ETL/ELT, Data Modeling, Power BI, Tableau, Streamlit, Power Query, Power Pivot

PROFESSIONAL EXPERIENCE

AI Engineer / Data Science Intern | ProHealth Home Health and Hospice | Atlanta, GA

Jun 2025 – May 2026

- Built an agentic text-to-SQL pipeline on AWS using Python, LangGraph, MCP tooling, document knowledge retrieval, DynamoDB for session memory and self-correcting loops, cutting administrator lookup time by 90%.
- Evaluated the agentic pipeline on a 100+ question set through Postman runs with RAGAS and human review, fixing retrieval inconsistency with AWS Opensearch hybrid retrieval and reranking to reach 92.6% context precision and 98% faithfulness.
- Engineered an ETL pipeline using Python, Apache Spark and SQL to ingest, transform and consolidate patient, financial and operational data into 30+ reusable views, replacing manual processing time by 45%.
- Retrained a readmission ML model on AWS Sagemaker using Python, scikit-learn and clinical feature engineering, running 30+ experiments through MLflow and cutting reamission cases by 10% (~\$15K savings).
- Architected three LangChain pipelines with parameterized SQL and deterministic chunking across predefined data points, giving finance and admin teams reproducible, auditable root-cause reports using LLMs, saving \$80K annually.
- Implemented LLMops, automated evaluations, prompt versioning, HIPAA compliance, guardrails, and observability using MLflow, LangSmith and OpenTelemetry, lowering AI production issues by 24%.
- Deployed AI Inference services on AWS using docker and Kubernetes, automating releases with GitHub Actions CI/CD and cutting model-serving costs by 17% via Redis caching and routing optimization.

Data Science Research Assistant | Syracuse University | Syracuse, NY

Jan 2025 – May 2025

- Fine-tuned a RoBERTa-based sentiment classification model using PyTorch and Hugging Face Transformers on 7,000+ survey responses, applying class-weight optimization and stratified k-fold validation to improve labeling accuracy by 30%.
- Applied Sentence-BERT embeddings and HDBSCAN clustering to identify distinct AI-adoption behavior groups from survey data.
- Analyzed sentiment, AI-adoption patterns, and KPIs to derive actionable research insights.
- Developed Power BI visualizations and analytical summaries to communicate sentiment distributions, behavioral segments, and KPI trends to research stakeholders.

Data Scientist | IBM | Mumbai, India

Jan 2023 – Jun 2024

- Developed and validated machine learning models using Python, Scikit-learn, and Microsoft Azure, improving predictive performance by 18% through feature selection, hyperparameter tuning, and cross-validation.
- Engineered large-scale data preprocessing pipelines using Python, SQL, and Azure data services, processing 1M+ records and reducing data preparation time by 35%.
- Designed optimized SQL queries and dimensional data models, reducing analytical query latency by 40% and supporting scalable ML and reporting workflows on Azure.
- Performed EDA, statistical analysis, and model validation using cross-validation and performance metrics, improving model selection accuracy by 20% and identifying key business drivers.
- Collaborated with business stakeholders and cross-functional teams to translate requirements into data and ML solutions, improving reporting efficiency by 30%.

PROJECTS

PrivateRAG | Python, Qdrant, Llama 3.1, TypeScript

Built a fully local RAG system for private-document Q&A with Qdrant vector indexing, metadata filtering, and Llama 3.1 inference, featuring source attribution and chunk-level traceability. | [Link](#)

AI Email Response System | Python, RAG, MCP, LLM-as-a-Judge

Developed an automated RAG-based email response workflow with policy checks, MCP-based routing, LLM-as-a-Judge validation, and human handoff for high-risk cases, reducing response wait time by 90%. | [Link](#)

EDUCATION

M.S. Information Systems (Data Science), Syracuse University — GPA: 3.86/4.00

Aug 2024 – May 2026

B.E. Information Technology, Thadomal Shahani Engineering College, Mumbai — GPA: 8.52/10

Aug 2020 – Jun 2024

CERTIFICATIONS

- AI Fluency** — Anthropic, 2026
- AWS Certified Machine Learning Engineer – Associate** — Amazon Web Services
- Generative AI with Large Language Models** — DeepLearning.AI / AWS